

CBSE 2027 Physics — Predicted Paper (Set 6 of 10)

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. Two point charges q_1 and q_2 are placed at a distance r apart in a vacuum. The electrostatic force between them is F . If a completely solid copper plate of thickness r is introduced to entirely fill the space between them, the new force will become: (a) F (b) $F/2$ (c) Zero (d) Infinite (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]**

Q2. A hollow spherical metal shell of radius **10 cm** is charged such that the electric potential on its outer surface is **5 V**. What will be the electric potential at the centre of the sphere? (a) Zero (b) **5 V** (c) **10 V** (d) **0.5 V** (*Chapter: Electrostatic Potential and Capacitance*) | **[Confidence: Very High]**

Q3. Kirchhoff's loop rule for any closed mesh of an electrical circuit is a direct mathematical consequence of the law of conservation of: (a) Charge (b) Energy (c) Linear Momentum (d) Angular Momentum (*Chapter: Current Electricity*) | **[Confidence: Very High]**

Q4. A charged particle enters a uniform magnetic field strictly perpendicular to the direction of the field. Which of the following physical quantities of the particle will remain constant during its motion? (a) Velocity (b) Momentum (c) Kinetic Energy (d) Acceleration (*Chapter: Moving Charges and Magnetism*) | **[Confidence: Very High]**

Q5. When a small piece of diamagnetic substance is placed in a non-uniform magnetic field, it tends to move: (a) from the weaker to the stronger part of the field. (b) from the stronger to the weaker part of the field. (c) perpendicular to the magnetic field. (d) nowhere, it remains perfectly at rest. (*Chapter: Magnetism and Matter*) | **[Confidence: High]**

Q6. The polarity of the induced emf in a closed loop is always such that it tends to produce a current which opposes the change in magnetic flux that produces it. This statement is known as: (a) Faraday's Law (b) Ampere's Law (c) Lenz's Law (d) Biot-Savart Law (*Chapter: Electromagnetic Induction*) | **[Confidence: Very High]**

Q7. In a purely inductive AC circuit, the alternating current: (a) leads the alternating voltage by a phase angle of $\pi/2$. (b) lags behind the alternating voltage by a phase angle of $\pi/2$. (c) is in same phase with the alternating voltage. (d) lags behind the alternating voltage by a phase angle of π . (*Chapter: Alternating Current*) | **[Confidence: Very High]**

Q8. Which of the following electromagnetic waves has the maximum wavelength? (a) X-rays (b) Ultraviolet rays (c) Infrared rays (d) Radio waves (*Chapter: Electromagnetic Waves*) | **[Confidence: High]**

Q9. When a monochromatic ray of light travels from an optically rarer medium to an optically denser medium, which of its following characteristics remains entirely unchanged? (a)

Velocity (b) Wavelength (c) Frequency (d) Amplitude (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

Q10. In Young's Double Slit Experiment (YDSE), if the two slits are illuminated by two completely independent identical sodium lamps, the interference pattern on the screen will: (a) become perfectly sharp and bright. (b) shift towards the right. (c) shift towards the left. (d) not be formed at all. (*Chapter: Wave Optics*) | **[Confidence: Very High]**

Q11. The momentum of a photon associated with light of wavelength λ is given by: (a) $h\lambda$ (b) h/λ (c) λ/h (d) hc/λ (*Chapter: Dual Nature of Radiation & Matter*) | **[Confidence: Very High]**

Q12. Nuclear forces are inherently: (a) long-range, attractive, and charge-dependent. (b) short-range, attractive, and charge-independent. (c) short-range, repulsive, and charge-independent. (d) long-range, repulsive, and charge-dependent. (*Chapter: Nuclei*) | **[Confidence: Very High]**

Q13. Assertion (A): Two electric lines of force can never intersect each other. **Reason (R):** If they intersected, there would be two different directions of the electric field at the point of intersection, which is physically impossible. (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]**

Q14. Assertion (A): The resistance of an ideal voltmeter should be infinite. **Reason (R):** An ideal voltmeter is connected in parallel and should not draw any current from the circuit so it can measure the true potential difference. (*Chapter: Current Electricity*) | **[Confidence: High]**

Q15. Assertion (A): In the photoelectric effect, the stopping potential strongly depends on the frequency of the incident light. **Reason (R):** The stopping potential is entirely independent of the intensity of the incident light, provided the frequency remains constant. (*Chapter: Dual Nature of Radiation & Matter*) | **[Confidence: High]**

Q16. Assertion (A): Silicon is generally preferred over Germanium for fabricating semiconductor devices. **Reason (R):** The energy band gap of Silicon is greater than that of Germanium, granting it better thermal stability at room temperature. (*Chapter: Semiconductor Electronics*) | **[Confidence: Medium]**

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. Define electric dipole moment. Sketch the pattern of electric field lines for an electric dipole (a system of two equal and opposite point charges separated by a small distance). (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]**

Q18. A moving coil galvanometer of resistance $50\ \Omega$ shows full-scale deflection for a current of $5\ \text{mA}$. How will you successfully convert it into an ammeter of range 0 to $2\ \text{A}$? (Show calculations). (*Chapter: Moving Charges and Magnetism*) | **[Confidence: Very High]**

Q19. Name the specific electromagnetic waves that are used in (i) cellular phones for communication, and (ii) water purifiers to kill germs. Write their approximate frequency

ranges. (*Chapter: Electromagnetic Waves*) | **[Confidence: High]**

Q20. An object is placed exactly at the principal focus of a concave lens. Where will the image be formed? Support your answer with a neat, labelled ray diagram. (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Medium]**

Q21. State Bohr's quantization condition for the angular momentum of an electron. Use it to calculate the angular momentum of an electron revolving in the 3^{rd} orbit of a hydrogen atom (in terms of Planck's constant h). (*Chapter: Atoms*) | **[Confidence: Very High]**

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. Derive a mathematical expression for the total electrostatic potential energy of a system of three point charges q_1 , q_2 , and q_3 located at position vectors $\vec{r}_1$, $\vec{r}_2$, and $\vec{r}_3$ respectively, in the absence of any external electric field. (*Chapter: Electrostatic Potential and Capacitance*) | **[Confidence: High]**

Q23. A cell of unknown emf $\mathcal{E}$ and internal resistance r is connected across two external resistors R_1 and R_2 . When R_1 and R_2 are connected in series across the cell, the current drawn is I_1 . When they are connected in parallel across the same cell, the current drawn is I_2 . Deduce expressions for the emf $\mathcal{E}$ and internal resistance r in terms of R_1 , R_2 , I_1 , and I_2 . (*Chapter: Current Electricity*) | **[Confidence: Medium]**

Q24. State Ampere's Circuital Law. Applying this law, find an expression for the magnitude of the magnetic field at a perpendicular distance r from a long, straight, infinitely thin current-carrying wire. (*Chapter: Moving Charges and Magnetism*) | **[Confidence: Very High]**

Q25. A coil of self-inductance L is connected in series with a glowing bulb B and an AC source. (a) What will happen to the brightness of the bulb if a soft iron core is slowly inserted into the inductor coil? Give a scientific reason. (b) What would happen if the AC source was replaced by a pure DC battery and the iron core was inserted? (*Chapter: Electromagnetic Induction / Alternating Current*) | **[Confidence: Very High]**

Q26. Draw a neat ray diagram showing the formation of an image by a concave mirror when an object is placed beyond the center of curvature (C). Using this diagram, derive the mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$. (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: High]**

Q27. An electron and a proton are accelerated from rest through the exact same potential difference V . (a) Find the ratio of their associated de-Broglie wavelengths ($\lambda_e : \lambda_p$). (b) Which of the two particles will acquire a higher kinetic energy? Justify your answer. (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: Very High]**

Q28. Clearly distinguish between the processes of nuclear fission and nuclear fusion (provide at least two differences). Why is it extremely difficult to carry out controlled nuclear fusion in a laboratory setting? Explain briefly using the concept of the Coulomb barrier. (*Chapter: Nuclei*) | **[Confidence: Medium]**

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.

Q29. Case Study 1: Extrinsic Semiconductors and Doping *Passage:* Pure semiconductors like Silicon and Germanium have a very low electrical conductivity at room temperature, making them impractical for creating electronic devices. To significantly improve their conductivity, a carefully controlled amount of suitable impurity is added to the pure crystal in a process called doping. Depending on whether the impurity added is pentavalent (like Phosphorus or Arsenic) or trivalent (like Boron or Indium), the resulting extrinsic semiconductor becomes *n*-type or *p*-type, vastly altering the concentration of majority and minority charge carriers. (a) Name the majority charge carriers in a *p*-type semiconductor. (1 mark) (b) What is the overall net electrical charge of an *n*-type semiconductor crystal? (1 mark) (c) In a doped semiconductor, the electron number density is $5 \times 10^{20} \text{ m}^{-3}$ and the hole number density is $2 \times 10^9 \text{ m}^{-3}$. Identify the type of semiconductor (*n*-type or *p*-type) and calculate the intrinsic carrier concentration (n_i). **OR** Why exactly does doping a pure Silicon crystal with a pentavalent impurity make it an *n*-type semiconductor? Explain briefly. (2 marks) (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]**

Q30. Case Study 2: Drift Velocity and Electron Mobility *Passage:* In a metallic conductor, free electrons move randomly in all directions like gas molecules in a container. Because their motion is random, their average velocity is zero, and they produce no net current. However, when a potential difference is applied across the ends of the conductor, an electric field is established inside it. The free electrons experience an electric force and begin to drift towards the positive terminal with a small, steady average velocity superimposed on their random thermal motion. This is called drift velocity. (a) Define the term 'relaxation time' in the context of electron drift in conductors. (1 mark) (b) How does the relaxation time of free electrons in a metal vary with an increase in temperature? (1 mark) (c) A given copper wire of length L and radius r is stretched to strictly double its original length. If it is then connected across the exact same voltage source V , how will the drift velocity of electrons in the wire be affected? **OR** Deduce Ohm's Law vectorially ($\vec{J} = \sigma \vec{E}$) utilizing the relationship between current density, drift velocity, and relaxation time. (2 marks) (*Chapter: Current Electricity*) | **[Confidence: Very High]**

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) Two capacitors of capacitances C_1 and C_2 are connected in series. Derive the expression for the equivalent capacitance C_{eq} of the combination. **(b)** A 600 pF capacitor is fully charged by a 200 V DC supply. It is then disconnected from the supply and connected in parallel to another completely uncharged 600 pF capacitor. Calculate the total electrostatic energy lost in the process of charge sharing. Where does this lost energy go? **OR (a)** Derive

an expression for the capacitance of a parallel plate capacitor when a conducting slab of thickness ' t ' ($t < d$) is introduced symmetrically between its plates, where ' d ' is the original separation between the plates. **(b)** A spherical conducting shell of inner radius R_1 and outer radius R_2 has a total net charge $+Q$ residing on it. A separate point charge $+q$ is placed exactly at the center of this shell. Determine the surface charge density on the (i) inner surface, and (ii) outer surface of the shell. (*Chapter: Electrostatic Potential and Capacitance / Electric Charges and Fields*) | **[Confidence: Very High]**

Q32. (a) A series LCR circuit is connected to an alternating voltage source $V = V_0 \sin(\omega t)$. Using a phasor diagram, derive the mathematical expression for the total impedance (Z) of the circuit. **(b)** Define the 'resonance frequency' of this circuit. Obtain its mathematical expression. **(c)** Draw a graph showing the variation of current amplitude (I_0) with the angular frequency (ω) of the AC source for two different values of resistance R_1 and R_2 (where $R_1 > R_2$). **OR (a)** Define motional electromotive force (EMF). **(b)** A straight conducting rod of length l is moving with a constant velocity v perpendicular to a uniform magnetic field B . Using the concept of the Lorentz magnetic force on free charge carriers, derive the expression for the motional EMF ($e = Blv$) induced across the ends of the rod. **(c)** A wheel with 10 metallic spokes, each 0.5 m long, is rotated with an angular speed of 120 rev/min in a plane normal to the horizontal component of the earth's magnetic field $H_E = 0.4 \text{ G}$ at a place. Calculate the EMF induced between the axle and the rim of the wheel. (*Chapter: Alternating Current / Electromagnetic Induction*) | **[Confidence: Very High]**

Q33. (a) State Huygens' principle of wave propagation. **(b)** Describe the diffraction pattern formed on a screen when a monochromatic source of light illuminates a single narrow slit of width ' a '. **(c)** Mathematically show that the linear width of the central maximum in the single-slit diffraction pattern is exactly twice as wide as the width of any of the secondary maxima. **OR (a)** Draw a neat, labelled ray diagram of a reflecting type telescope (Cassegrain telescope). **(b)** State two major advantages of a reflecting telescope over a standard refracting telescope. **(c)** In modern astronomy, the objective mirror of a telescope is designed to have a very large aperture compared to the eyepiece. Explain the two primary scientific reasons for this design choice. (*Chapter: Wave Optics / Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

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